Minimum Continuous Stable Flow (MCSF) method and calculation

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1. Purpose

This document establishes the criteria to define a Minimum Continuous Stable Flow (MCSF) for a centrifugal pump, i.e. the minimum operating capacity for which it is possible to guarantee pump vibration level on bearing housing within limits as per Table 8 of API 610 Std.

An additional method for MCSF estimation of non-API low energy (refer to SOP-489) vertical pumps is also proposed.

2. Scope

This procedure is applicable to both single and double suction impeller, single- or multi-stage pumps.

3. Definitions

BEP Best Efficiency Point

MCSF Minimum Continuous Stable Flow

NPSHr Net Positive Suction Head required

SSS Suction Specific Speed

4. General Procedure - Description

The definition of pump MCSF can pass through a calculation done using the MOD-414 spreadsheet as starting value.

This method (defined by Bradshaw S.) takes into account of the following pump features:

- Single/double suction 1st stage impeller
- Rotational speed
- BEP head per stage at maximum diameter
- BEP flow at maximum and minimum diameter
- BEP power at maximum diameter
- BEP NPSHr at maximum diameter (SSS)

These are also the information necessary in the calculation module.

5. Methodology, analysis and interpretation of results

Once filled the input data (yellow cells of Figure 1), MCSF is evaluated according to the graph for Bradshaw's method which is shown in Figure 2. The Excel file is available also in metric units and then input data are converted to US units.

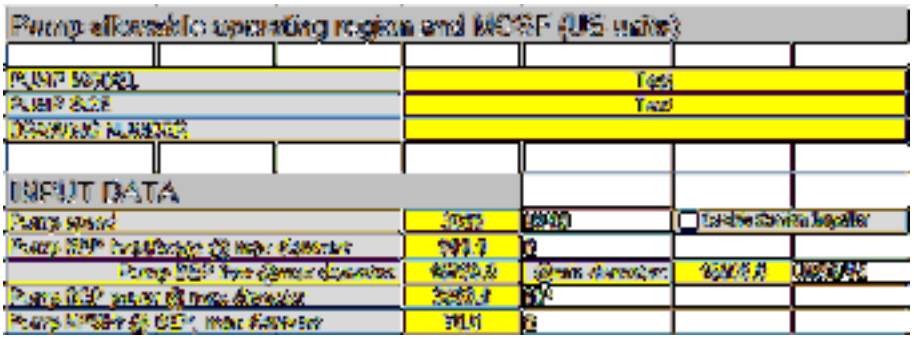

Figure 1: Input data for MCSF calculation (Bradshaw's method)

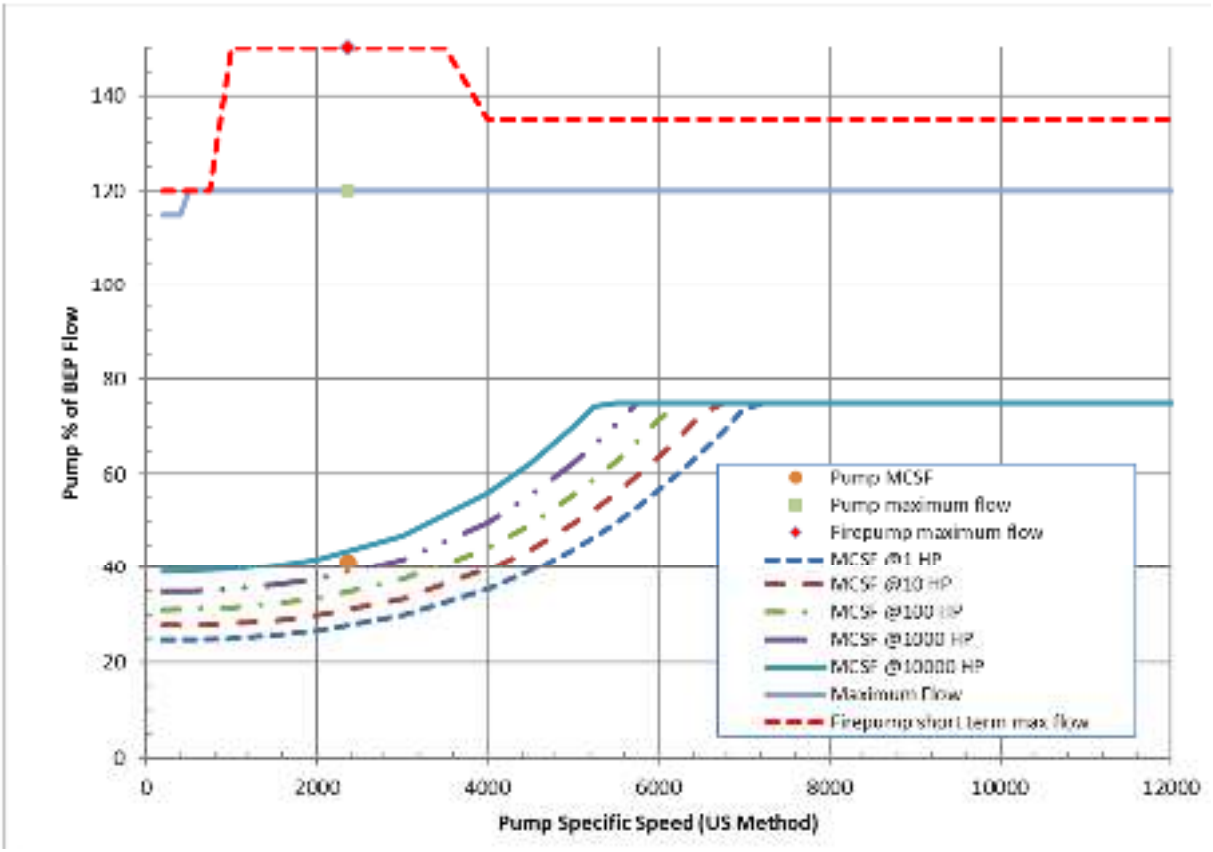

Figure 2: Graph for MCSF evaluation according to Bradshaw method

In the output data (see Figure 3) it is possible to see the three different contributions on which MCSF calculation is based on:

- Pump specific speed and power
- High speed operation
- High SSS

Some limitations on this calculation method shall be considered:

- Pumps with a pronounced saddle on head-capacity curve (or instable low flow), MCSF should always be set so that the performance curve is continuously rising up to minimum flow.
- Some older pump designs may require even more restrictive flow limits than those defined here.
- Unstable NPSHr curves at low flow or asymptotic NPSHr at high flow may further limit operating range.

However, resulting MCSF is a first attempt value and shall be checked against experimental test data of the considered pump.

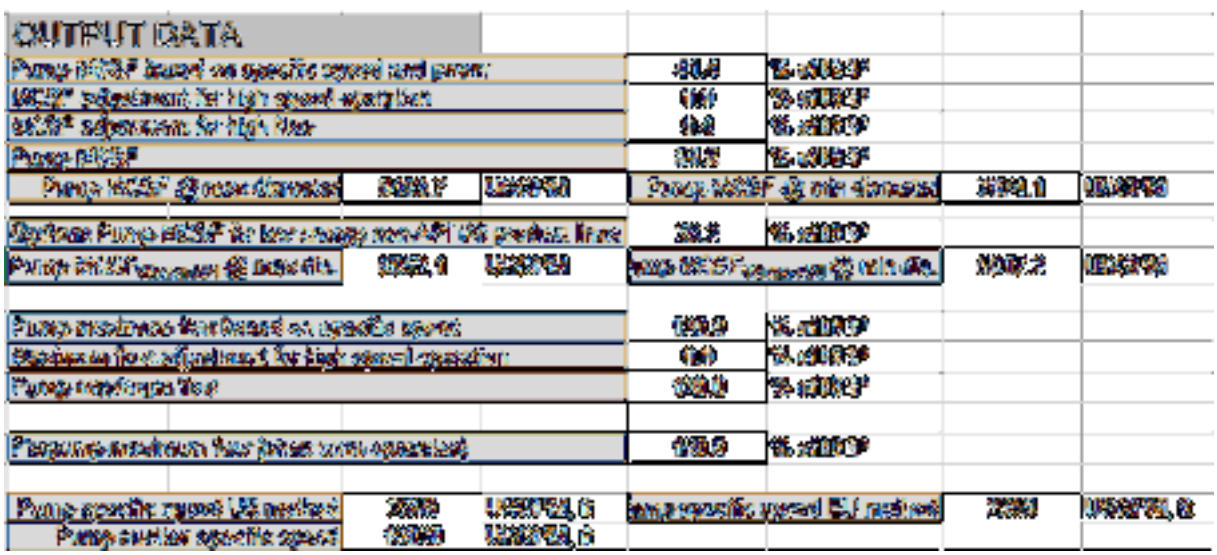

Figure 3: Output data of Bradshaw's method for MCSF calculation

For low energy vertical shaft product lines (e.g. Floway legacy), not subjected to API requirements in terms of vibration limits, it is possible to use the following corrective equation in order to estimate the MCSF, because, for this kind of pumps, typical values are limited max to roughly 30-40% of BEP:

$$MCSF_{VS\ non-API} = 0.3 + 0.3(MCSF_{API} - 0.3)$$

This value is optionally indicated in the standard calculation spreadsheet (7th line of Figure 3). A comparison of MCSF values calculated by the two formulations for all Floway legacy hydraulics is shown in Figure 4, where it is also possible to see that the correction works well to find values more typical of similar pumps.

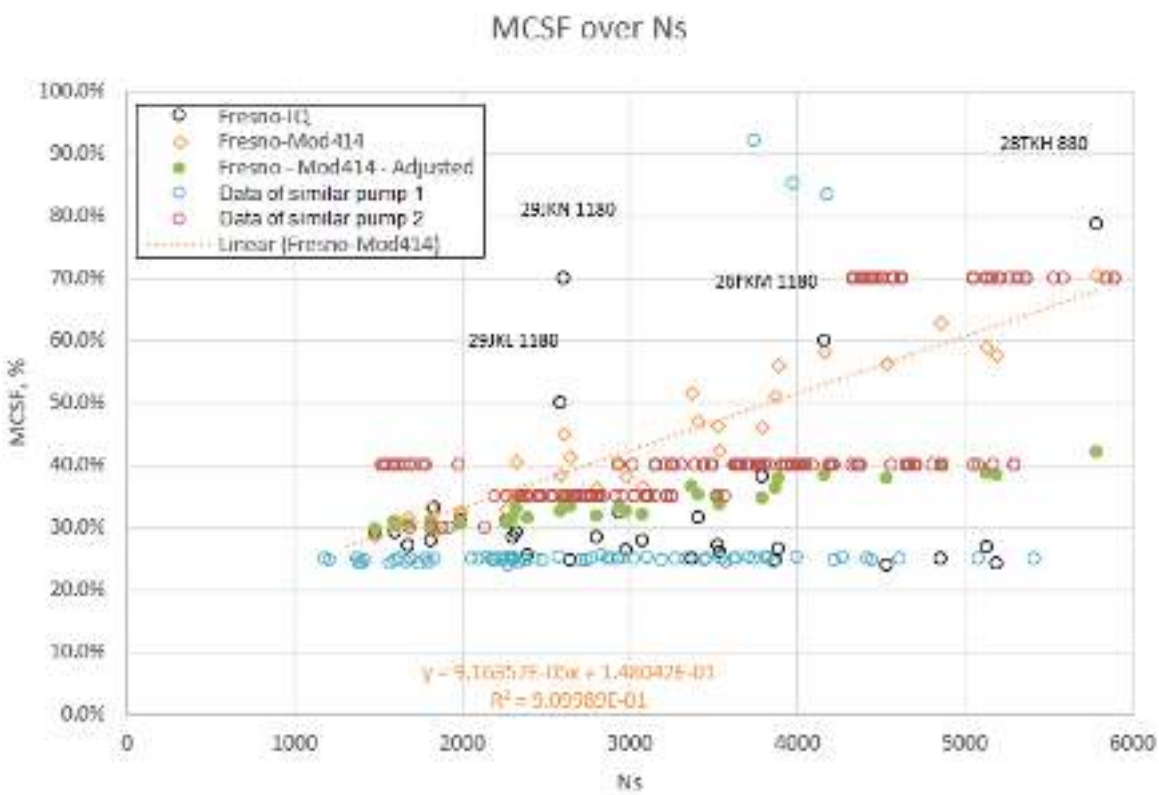

Figure 4. Comparison of the two methods of MCSF calculation for Floway legacy hydraulics and typical values of MCSF used for similar pumps

6. Acceptance Criteria

The definition of the Minimum Continuous Stable can be done using Bradshaw method and then it is verified testing the pump and measuring the vibration level at different capacities checking that the limits stated on Table 8 of API 610 Std. are not exceed.

Moreover, also temperature on bearing housing (oil or bearing metal temperature) shall be monitored to be within limits prescribed by API 610 Std. and indicating an acceptable level of thrust. In fact, at maximum loads (which can occur at MCSF) a rating life of at least 16000 hours shall also be guaranteed for rolling element bearings.

For non-API low energy vertical shaft pumps, lower values of MCSF are allowed also because higher vibration levels are acceptable.

7. References

- *API STANDARD 610, XII Edition, January 2021*
- *Mod.414 – Minimum Continuous Stable Flow spreadsheet*
- *SOP-489 - High-Medium-Low Energy pumps categorization*